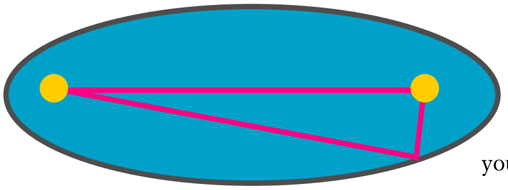
Planet Earth travels along an **ellipse** as it orbits the Sun



To draw an ellipse, loop a circle of string around two pins and pull the string with a pencil as you draw the curve.

WHAT'S AN ELLIPSE?

An ellipse looks like a squashed circle, but we can describe it more precisely by using maths. Inside an ellipse are two points called foci. The ellipse is the line made by all the points whose combined distance from the two foci is the same. Many people think planets orbit the Sun in circles, but in fact they travel along ellipses. The Sun is at one of the two foci of each planet's orbit. The other focus is just empty space.

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Galileo found a link between cannon balls and ***square numbers***. Whatever distance the ball falls in the first unit of time, by the second it will have fallen ***four times*** as far, and by the third, ***nine times*** as far. So the distance the ball falls increases in ratio with the square of the time.

Moon fall into the Earth?

Newton explained that the Moon is accelerating towards Earth, but it is going sideways precisely fast enough to keep it in orbit. By combining the work of Galileo and Kepler, Newton discovered how the force of gravity works.

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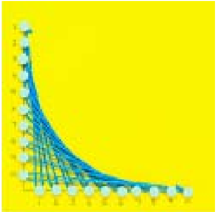
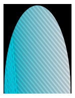
Curves from lines

To prove his theory of gravity, Isaac Newton invented a new branch of maths. He called it "fluxions", but we now call it calculus. Calculus is great for sums where something keeps changing, like the speed of an accelerating rocket. A graph would show this as a curve. By using calculus, we treat the curve as an infinite number of straight lines.



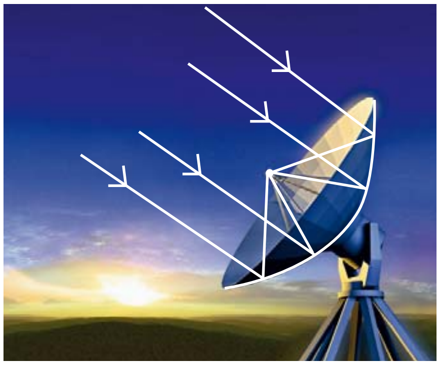
A cunning trick

The ancient Greeks didn't have calculus. Even so, Archimedes proved that the area under a parabola is two-thirds of the rectangle around it. How did he do it? He drew them on parchment, cut them out, and weighed them.



Useful parabolas

Parabolic mirrors reflect light to a point called a focus. Satellite dishes use this principle to collect faint signals from satellites and concentrate them onto a detector.



Radio telescopes use parabolic dishes to collect very faint signals from outer space.